

Assessing Efficacy of Gain-Framed and Loss-Framed Reminder Emails to Improve Statin Medication Adherence in Older Medicare Patients

Andrea Yeung, BA¹; Max Hukill, AB¹; Jonathan Alkalai, BS¹; Mandisa Keswa, BA¹; Isabel Arana, BS^{1*}; Kavenpreet Bal, MSc^{1*}; Iris Deng, BS^{1*}; Alex Gomez-Lara, BA, BS^{1*}; Mohamed Hamideh, BS^{1*}; Jorge Mercado, MD²; Erlinda Crisolo²; Bethелеhem Teshome²; Robyn Glezer, DO^{1,2}

1. Kaiser Permanente Bernard J Tyson School of Medicine; Pasadena, CA. 2. Kaiser Permanente Panorama City, Panorama City, CA.
* Denotes authors with equal contribution



Introduction

- Statins decrease risk of cardiovascular events, especially in older adults
- However, nearly half of patients aged 65 and older discontinue or inconsistently fill their prescriptions within the first year
- Personalized email outreach has proven efficacy in the primary care setting
- Framing of health messages (gain vs. loss) can meaningfully influence patients’ intentions and actions

Objective/Research Question

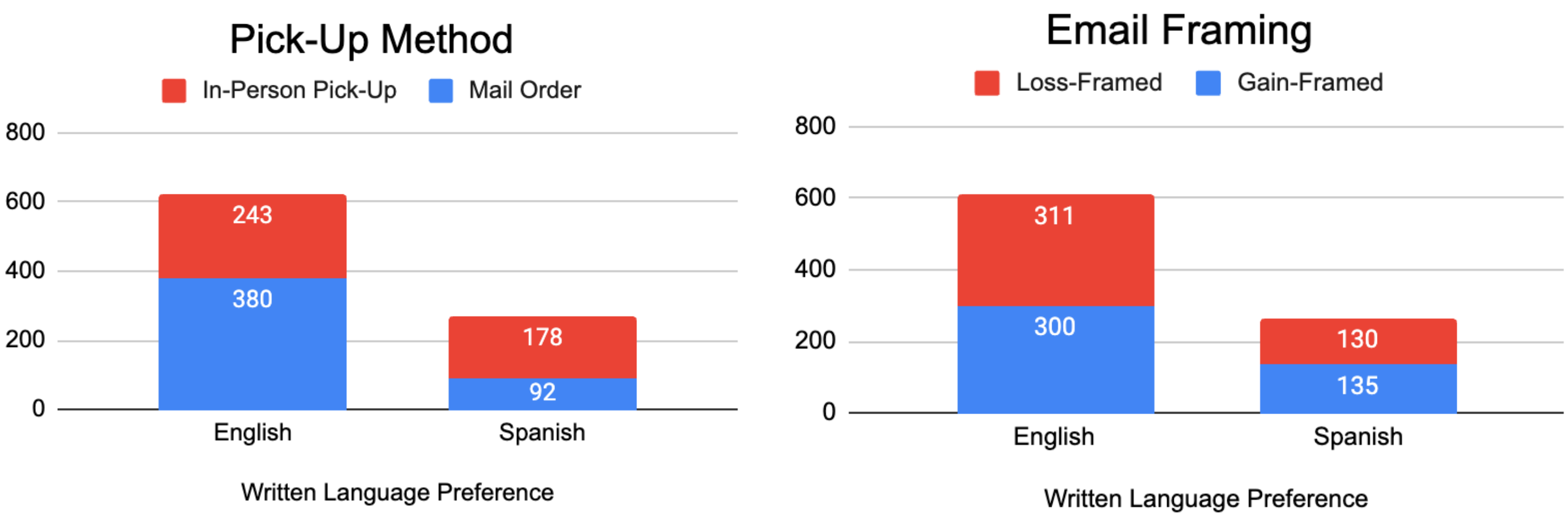
- To evaluate the effects of gain- and loss-framed personalized emails on statin adherence in the Panorama City region of Kaiser Permanente Southern California (KPSC)

Methods

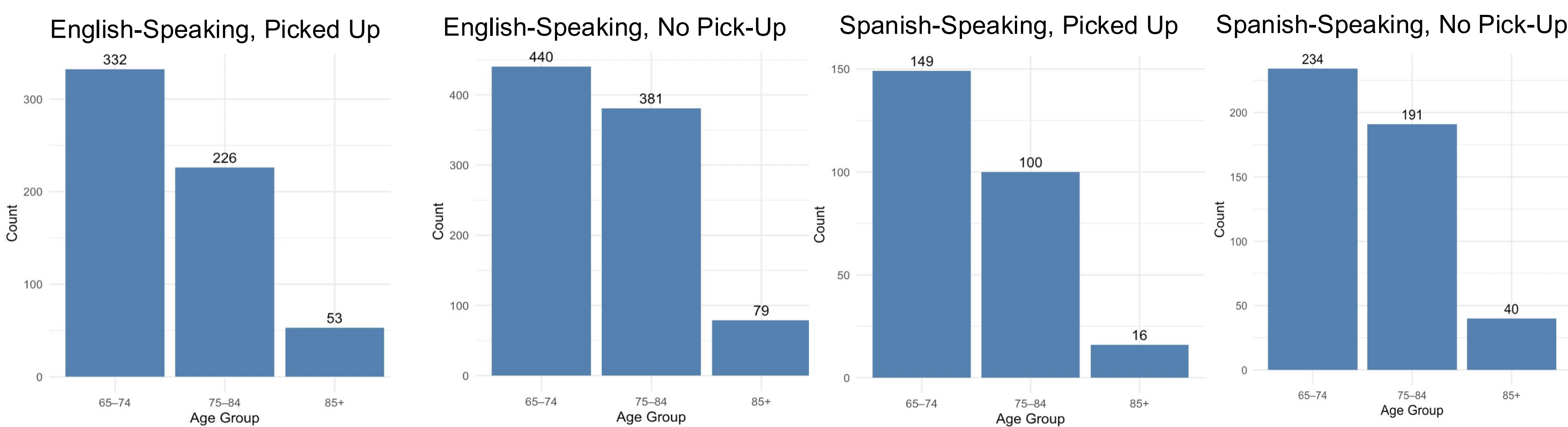
- **Population:** Medicare-insured patients aged ≥65 with low statin adherence who were English or Spanish speaking and active on kp.org
- **Intervention:** personalized gain-/loss-framed emails with ASCVD risk score and actionable links.
 - Gain-framed: positive outcomes, improved health
 - Loss-framed: negative outcomes, increased risk
- **Outcomes:** statin medication pick-up during the study period, whether in-person or mail order

	English-speaking		Spanish-speaking		Overall	
Total # patients	1511		730		2241	
Age (n, %)						
65–74	772,	51.09%	383,	52.47%	1155,	51.54%
75–84	607,	40.17%	291,	39.86%	898,	40.07%
85+	132,	8.73%	56,	7.67%	188,	8.39%
Gender/sex						
F	784,	51.89%	435,	59.59%	1219,	54.40%
M	727,	48.11%	295,	40.41%	1022,	45.60%
Race/ethnicity						
AAPI	328,	21.71%	0,	0%	328,	14.64%
American Indian / Alaska Native	7,	0.46%	1,	0.14%	8,	0.36%
Black	89,	5.89%	0,	0%	89,	3.97%
White	623,	41.23%	0,	0%	623,	27.80%
Latine	299,	19.79%	688,	94.25%	987,	44.04%
Multiracial	107,	7.08%	15,	2.05%	122,	5.44%
Other	58,	3.84%	26,	3.56%	84,	3.75%
Comorbidities						
Hypertension	1122,	74.26%	517,	70.82%	1639,	73.14%
T2DM	638,	42.22%	323,	44.25%	961,	41.88%
CKD	374,	24.75%	171,	23.42%	545,	24.32%
Smoking Status						
Non-Smoker	883,	58.44%	475,	65.07%	1358,	60.60%
Current Smoker	72,	4.77%	29,	3.97%	101,	4.51%
Quit	511,	33.82%	198,	27.12%	709,	31.64%
10-year ASCVD Flag (n, %)	152,	10.06%	75,	10.27%	227,	10.13%
PDC					0.67	

Table 1. Baseline demographics for all patients included in our study (Medicare, 65+, low adherence, active on kp.org)



Figures 1A, 1B: Medication pickups in English- and Spanish-speaking patients, by pick-up method and email framing.



Figures 2A-D: Age distribution of English- and Spanish-speaking patients who received emails in our study, separated by whether they picked up their medication or not

Results

- Among 2241 patients who received an email, 39.1% (n=876) of patients picked up a statin prescription during the 8-week effect period
- Email framing showed minimal differences in medication pickup:
 - English-speaking patients: 39.7% (gain) vs. 41.1% (loss)
 - Spanish-speaking patients: 37.0% (gain) vs. 35.6% (loss)
- Language preference influenced pickup modality:
 - English-speaking patients: 61% used mail-order
 - Spanish-speaking patients: 34% used mail-order
- The most common statin dispensed was atorvastatin (60%), followed by rosuvastatin (26%)
- The age distribution of those who picked up meds, compared to those who did not pick up meds, slightly favored younger patients in both English- and Spanish-speaking patients

Conclusion & Discussion

- Email framing did not seem to affect statin medication adherence in this study population and time frame
- There may be different preferences in mail-order vs in-person pick-up for English vs. Spanish speaking patients. Future research should explore whether sociocultural factors affect method of medication pickup
- A greater proportion of patients who picked up their medications was in the youngest age range (65-74), suggesting that technological literacy may impact the effectiveness of an email intervention
- Limitations:
 - Short study period - could not determine if emails were read
 - Contamination bias – cannot isolate from other events
 - Information bias – statin pick-up as a proxy for adherence

Acknowledgements

We thank Dr. Robyn Glezer, Dr. Elian Paiuk, Dr. Sabreena Arif, Bethелеhem Teshome, and the rest of the Kaiser Permanente Panorama City Area for their help and mentorship.